

Case Study 1. E-commerce funnel analysis

Main question

Where is the largest gap between product, cart and purchase activity, and which segments account for most of that gap?

Data and approach

I used the theLook eCommerce events dataset in BigQuery and built the visualizations in Power BI. The analysis compares counts of product, cart and purchase events, then breaks the product-to-purchase gap down by geography.

Method note: The dataset is counted at event level. The percentages in the first chart are event-count ratios, not user-level conversion rates, because the query does not connect ordered events to the same user or session.

Figure 1. Basic e-commerce funnel: product → cart → purchase.

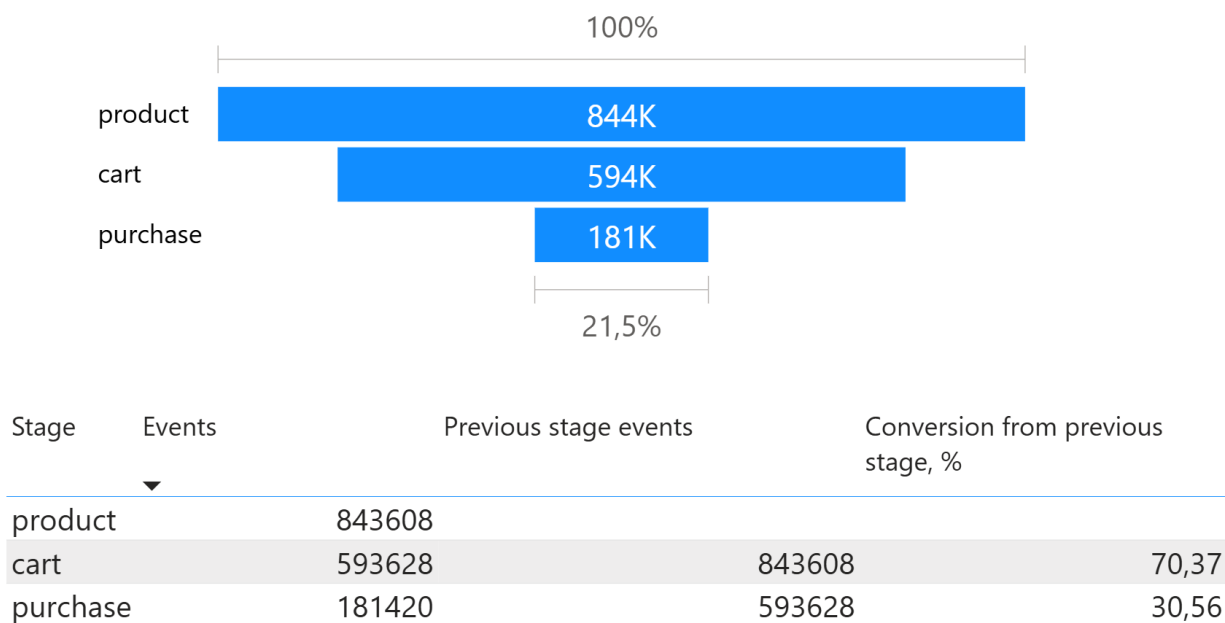


Figure 1. Product, cart and purchase event counts.

Result

The dataset contains 843,608 product events, 593,628 cart events and 181,420 purchase events. The largest count difference is between cart and purchase: 412,208 events. This makes the checkout stage the first place to investigate. It does not mean that 412,208 unique users abandoned their carts.

The browser and traffic-source splits were very similar, so they do not explain the gap and are not useful for prioritizing the next investigation.

Country-level results

Top 10 countries by total loss

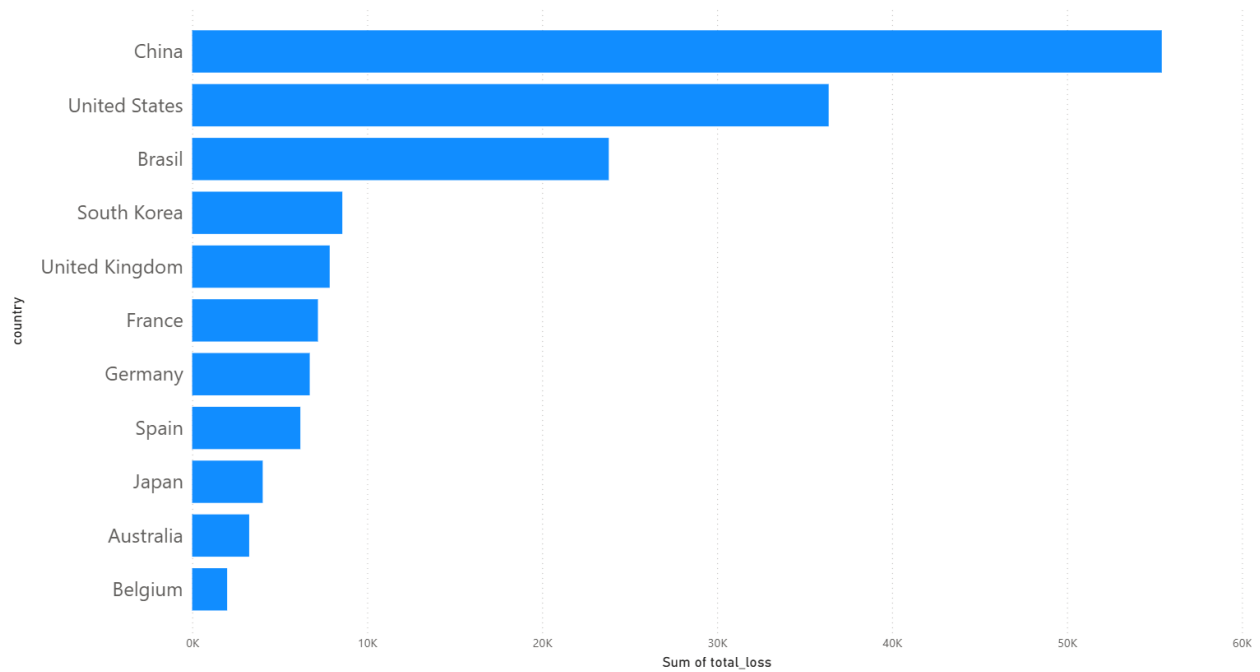


Figure 2. Top countries by total loss.

In this report, total_loss is the difference between product events and purchase events. China has the largest absolute gap, followed by the United States and Brazil.

This is mainly a volume ranking. Large markets naturally produce larger absolute gaps even when their event-count ratios are close to the overall average. The chart is useful for deciding where to inspect checkout data first, but it should not be read as a ranking of traffic quality or true user conversion.

City-level results

Cities by total loss

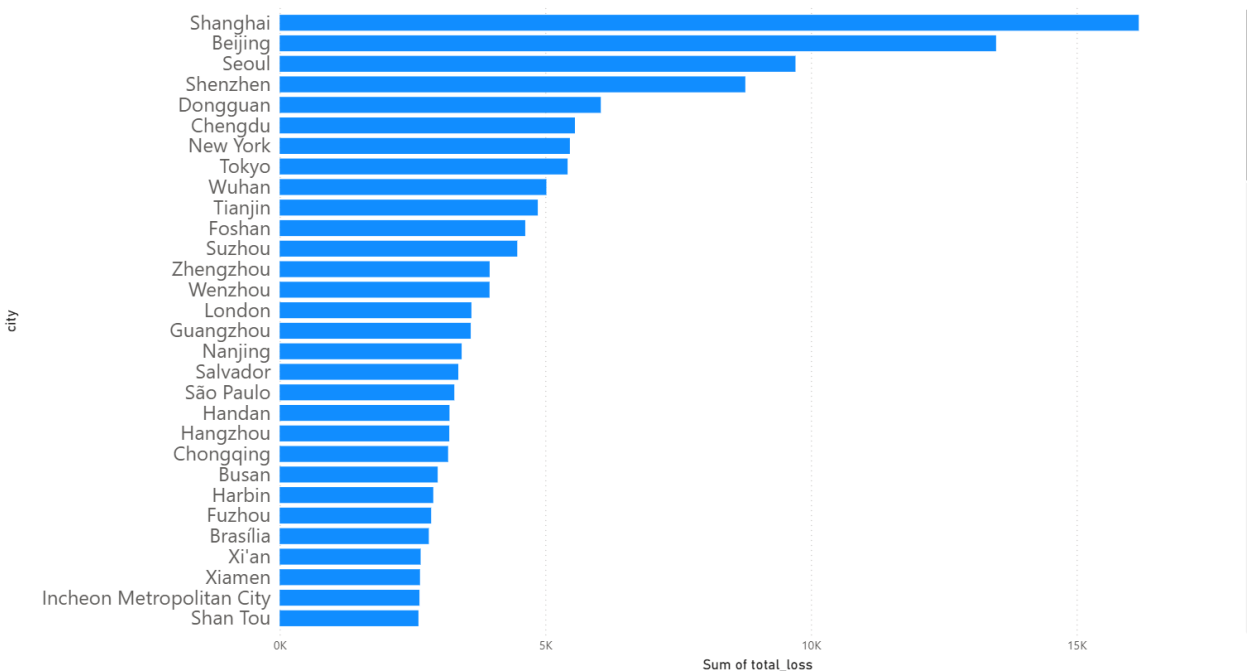


Figure 3. Cities by total loss.

The city view shows the same pattern in more detail. Shanghai and Beijing have the largest absolute gaps, followed by Seoul and Shenzhen. These cities are the best starting points because they have the largest product-to-purchase event gaps in absolute terms.

The result does not show that location causes the drop. It only narrows the next investigation to a smaller set of high-volume segments.

Volume and event-count ratio

City volume vs conversion

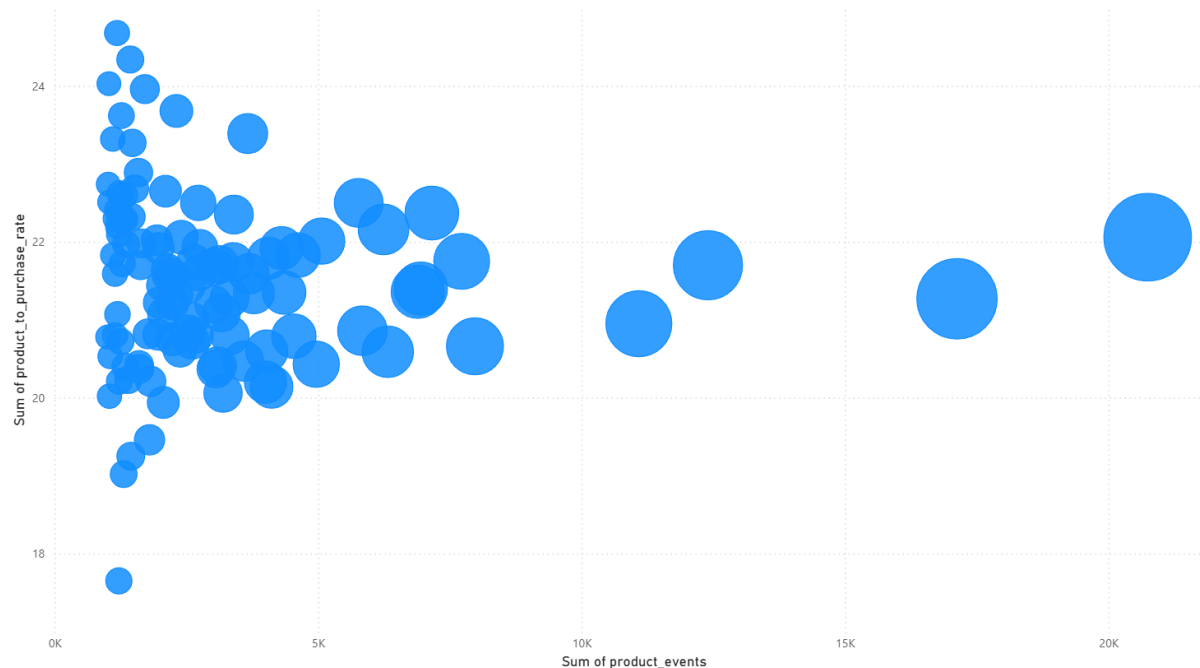


Figure 4. City volume vs conversion.

Most cities are grouped within a fairly narrow product-to-purchase event-count range. The largest cities do not appear to have sharply worse ratios than the rest. Their high total loss is driven mainly by scale.

Some low-volume cities show more extreme ratios, but they account for fewer events and are more sensitive to small-sample variation. They are less useful as the first priority for investigation.

Conclusion

The largest event-count gap appears after cart, and most of the absolute gap is concentrated in high-volume countries and cities. The current analysis shows where to investigate, but not why the gap occurs.

Next steps

Rebuild the funnel by user or session, keep the steps in chronological order and define a conversion window. Then add checkout-level events such as checkout start, shipping selection, payment attempt, payment error and purchase confirmation. After that, compare true conversion rates by geography, browser and traffic source using minimum sample-size filters.